SAFE & RESPONSIBLE AI SYSTEMS

Note: Week 10 = guest lectures 'AI Policies, AGI with Prof. David Krueger (Parts 1 & 2)'. No slides were in the project — reconstructed from the assignment Q&A + standard AI-safety/governance material. All 12 graded questions are covered.

1 Auditing & Demonstrating Safety

Black-box access is insufficient for rigorous auditing (Q1, select all):

- It **focuses only on final outputs** rather than internal mechanisms
- It **fails to reveal underlying problematic model behaviour**

(NOT: 'restricts to utilization of computation', NOT 'requires retraining weights from scratch'.)

Benchmarks are insufficient (Q4) for assessing dangerous capabilities because **benchmarks fail to capture emergent behaviours in a tool-based environment** — a model wired to tools/internet can do things no static benchmark tests.

Proving safety instead of third-party audits (Q5): developers can **provide a positive 'safety case' — an argument demonstrating that the system is safe** before deployment (burden of proof on the developer), rather than relying only on external audits.

Why safety fine-tuning fails to remove deep bias (Q9): superficial behaviour alignment — fine-tuning changes surface behaviour but leaves deep-seated societal biases embedded in the model.

2 Misalignment Failure Modes

Concept	Meaning
Goal misgeneralization (Q2)	The agent pursues a **different goal that is merely correlated** with the true reward function — learned the wrong goal that happened to match in training
Reward hacking	The agent **continuously optimizes an imperfect proxy goal**, worsening true outcomes — contrast: it games the given metric rather than pursuing a different goal
Persuasion optimization (Q10)	Optimizing AI for user persuasiveness is dangerous because it **creates incentives to manipulate human intent**

Assignment: goal misgeneralization = pursues a different but correlated goal (Q2); vs reward hacking = over-optimizing an imperfect proxy. Persuasion-optimization danger = incentives to manipulate human intent (Q10).

3 Why AGI / Agentic AI is Different

What makes highly agentic AI dangerous beyond current LLMs (Q3): direct interaction with external digital infrastructure (internet, tools, code, money) — not parameter count, token speed, or context-window size.

Cross-modal capability & AGI (Q12): cross-modal ability is considered a hallmark of AGI because **excluding human modalities creates a gap in real-world understanding** — true general intelligence must handle the same mix of modalities humans do.

Systemic risk of personalized assistants (Q6): highly capable assistants that adapt to a user's habits **put dangerous amounts of private information in the hands of a few big companies** — concentration of power & data.

4 Governance, Copyright & Green AI

Copyright / 'transformative fair use' (Q7): training on copyrighted data does NOT count as transformative fair use because **the resulting model directly competes economically with the creators** (undercuts the market for the original work).

Why single-nation regulation fails (Q8): individual nations may feel the need to bypass regulations and build dangerous AI systems anyway (a race dynamic) — so safety needs international coordination.

Upstream vs downstream regulation (Q11): regulate the **companies developing AI & hardware** (upstream) rather than the users (downstream), because **users face high market pressure to adopt AI systems** and can't opt out.

Assignment quick-key: Q1 outputs-only + hides problematic behaviour; Q2 different correlated goal; Q3 external digital infrastructure; Q4 emergent tool-based behaviours; Q5 positive safety case; Q6 data in few companies; Q7 competes economically with creators; Q8 nations bypass anyway; Q9 superficial behaviour alignment; Q10 manipulate human intent; Q11 users face market pressure; Q12 excluding modalities → real-world gap.